



ANDERSON JUNIOR COLLEGE
2013 JC 2 PRELIMINARY EXAMINATIONS

NAME: _____ **PDG:** _____ /12 **Register No:** _____

CHEMISTRY

Higher 1

Paper 2 Structured Questions

8872/02

16 September 2013

2 hours

Candidates answer Section A on the Question Paper

Additional Materials: Answer Paper
 Graph Paper
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use			
Paper 1	/ 30	Total	%
Q1			
Q2			
Q3			
Q4		Grade	
Q5			
Q6			
Q7			
Paper 2	/ 80		

This document consists of **15** printed pages.

- 1 The Pollutant Standards Index (PSI) is an air quality indicator. It is based on five pollutants: particulate matter (PM₁₀), sulfur dioxide, carbon monoxide, ozone and nitrogen dioxide.

To calculate the overall PSI, the PSI value is first calculated for each of the five pollutants, using the following table.

i	PSI value, P _i	Concentration, C _i				
		PM ₁₀ ($\mu\text{g m}^{-3}$)	SO ₂ ($\mu\text{g m}^{-3}$)	CO (mg m^{-3})	Ozone ($\mu\text{g m}^{-3}$)	NO ₂ ($\mu\text{g m}^{-3}$)
1	50	50	80	5.0	118	–
2	100	150	365	10.0	157	–
3	200	350	800	17.0	235	1130
4	300	420	1600	34.0	785	2260
5	400	500	2100	46.0	980	3000
6	500	600	2620	57.5	1180	3750

[1 $\mu\text{g} = 10^{-6}$ g; 1 mg = 10^{-3} g]

Given the concentration of a pollutant (with units as stated in the table above),

$$\text{PSI of pollutant} = \frac{P_{i+1} - P_i}{C_{i+1} - C_i} (\text{concentration of pollutant} - C_i) + 100,$$

where $C_{i+1} > \text{concentration of pollutant} > C_i$

The overall PSI is the maximum value out of the five calculated PSI pollutant values.

In Singapore, the 24-hr PSI is used by the National Environment Agency (NEA) to provide health advisory.

24-hr PSI	Healthy Persons	Elderly, Pregnant women, Children	Persons with chronic lung disease, heart disease, stroke
< 100	Normal activities		
101–200	Minimise prolonged or strenuous outdoor physical exertion.	Minimise prolonged outdoor activity.	Avoid all outdoor activities. If outdoor activity is unavoidable, wear N95 mask.
201–300	Avoid prolonged or strenuous outdoor physical exertion. If outdoor activity is unavoidable, wear N95 mask.	Avoid all outdoor activities.	
>300	Minimise all outdoor exposure. If outdoor activity is unavoidable, wear N95 mask.	If outdoor activity is unavoidable, wear N95 mask (for adults).	

- (a) (i) In a 1 m^3 sample of air, the mass of PM10 and carbon monoxide were found to be $320 \text{ } \mu\text{g}$ and 20 mg respectively. Calculate the PSI values for each of the above pollutants, leaving your answers to **3 significant figures**.

PM10: CO:

- (ii) What is the maximum concentration of carbon monoxide, in mg m^{-3} , allowed in a sample of air in order for its PSI value to not exceed 150?

CO: mg m^{-3}

- (iii) Given that the PSI value of sulfur dioxide, ozone and nitrogen dioxide are 150, 112 and 133 respectively, use your answer to (a)(i) to determine the overall PSI.

PSI:

- (iv) Assuming that the current overall PSI level is your answer in (a)(iii), what advice would you give to a 7-year-old who wants to play at the outdoor playground?

.....

.....

[5]

- (b) The amount of sulfur dioxide in a sample of air can be determined by first reacting it with sodium iodate, NaIO_3 , and then titrating the sulfuric acid formed with sodium hydroxide. Iodine is also formed together with sulfuric acid.

(i) Write an ionic equation for the reaction between sulfur dioxide and sodium iodate.

- (ii) When a 1 m^3 sample of air was bubbled through a solution of sodium iodate, the resulting solution was neutralised by 10.0 cm^3 of $0.005 \text{ mol dm}^{-3}$ sodium hydroxide solution.

Calculate the concentration of sulfur dioxide, in $\mu\text{g m}^{-3}$, in the sample of air.

Concentration of SO_2 : $\mu\text{g m}^{-3}$
[3]

- (c) Some countries have set limits for particulates in the air. For example, the European Union has a daily average (24-hour) limit of $180 \mu\text{g m}^{-3}$ for PM10. Cities that violate this daily limit face a hefty financial penalty that is calculated with consideration of many factors such as the severity the violation has on the ecosystem, duration of the non-compliance and country's GDP etc.

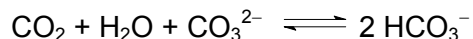
- (i) A collected sample of air from the German city of Leipzig contains 0.00002% by mass of PM10. Given that the density of air is 1 kg m^{-3} , calculate the concentration of PM10 in the sample collected.

- (ii) Hence, deduce whether the German city of Leipzig will be faced with any financial penalty.

.....
[2]
[Total: 10]

- 2 The burning of fossil fuels, which results in the production of CO_2 , have been a concern to the environment. It was known that ocean water has become 30% more acidic (in terms of H^+ concentration) over the last 150 years. When carbon dioxide reacts with water, carbonic acid, H_2CO_3 is formed and this causes acidification of water in oceans and affects marine life.

The following equilibrium in ocean water is established.



HCO_3^- and its conjugate base are the main components in the ocean that regulates its pH.

- (a) (i) A 100 g sample of ocean water saturated with pure carbon dioxide at atmospheric pressure contains 0.15 g of dissolved CO_2 . Calculate the concentration of dissolved CO_2 , in mol dm^{-3} , present in this sample.
[Assume that ocean water contains only CO_2 and water with a density of 1 g cm^{-3}]

$$[\text{CO}_2(\text{aq})] = \dots\dots\dots \text{mol dm}^{-3}$$

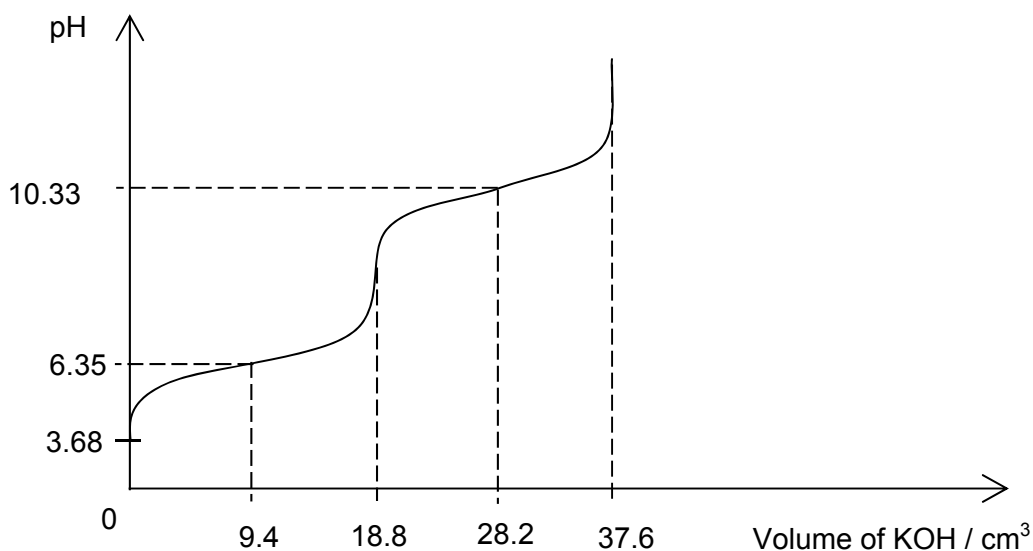
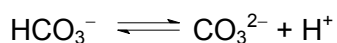
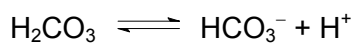
- (ii) If the pH of ocean water 150 years ago was 8.25, calculate its current pH.

- (iii) With the aid of two balanced equations, show how the pH of the ocean is regulated.

.....

[5]

- (b) In an experiment, 25 cm³ of 0.100 mol dm⁻³ H₂CO₃ is titrated with aqueous potassium hydroxide. The titration curve is shown below. The acid dissociation of H₂CO₃ is as shown in the following equations.



- (i) Justify that carbonic acid, H₂CO₃, is a weak acid with relevant calculations.

- (ii) Calculate the concentration of KOH, in mol dm⁻³, used in the titration.

- (iii) Suggest, with a reason, a suitable indicator for the first end point of this titration.

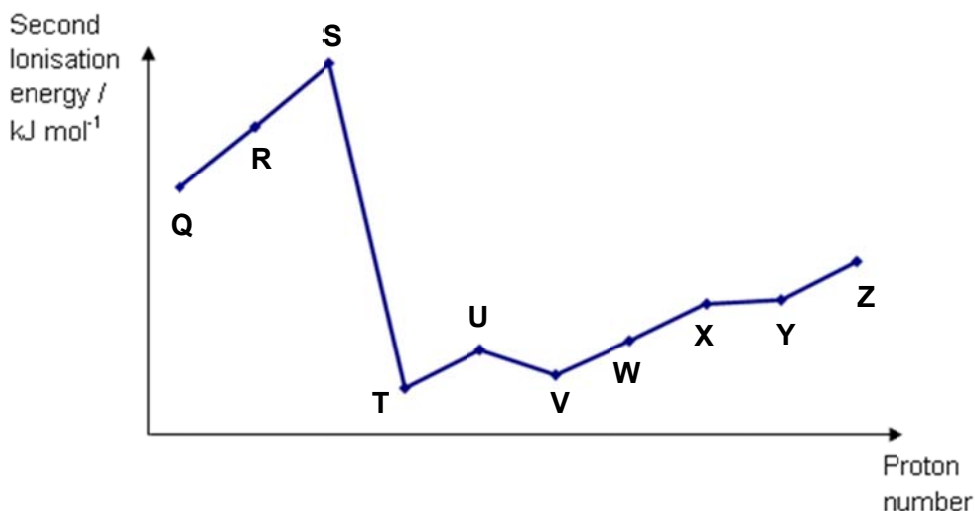
.....

.....

[5]
[Total: 10]

- 3 (a) In an analytical chemistry laboratory, the plasma torch is often used to ionise a variety of samples.

In one series of analysis, samples of unknown elements were passed through the plasma torch. The **second ionisation energies** of ten successive elements **Q** to **Z** in both Period 2 and 3 of the Periodic Table are recorded and plotted as shown below, with one of the elements being phosphorus.



- (i) Which one of the ten elements represents phosphorous?

.....

- (ii) Element **Q** and **Y** are in the same group. Using the electronic configurations of **Q⁺** and **Y⁺**, explain why the second ionisation energy of **Y** is found to be lower than that of **Q**.

Electronic configuration

Q⁺ **Y⁺**

.....

.....

.....

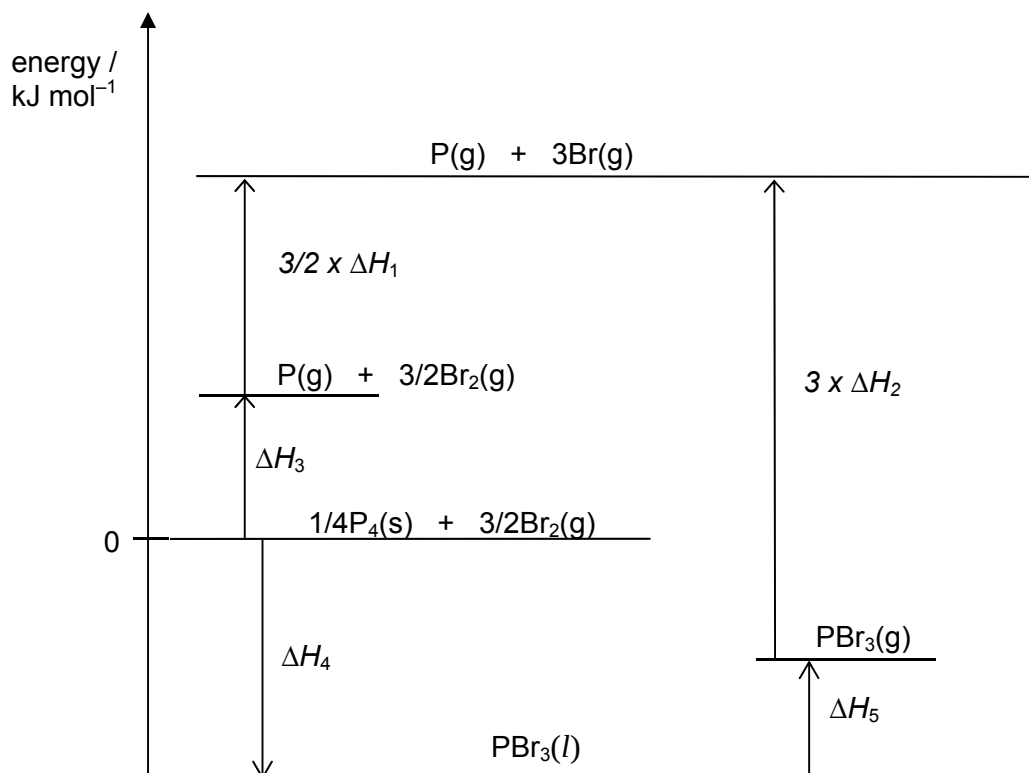
.....

- (iii) Suggest the formula of a compound that element **V** will form with chlorine.

Formula of compound:

[5]

- (b) Liquid phosphorous tribromide is prepared by the reaction of bromine with red phosphorus, P_4 . An energy level diagram which starts from $P_4(s)$ and $Br_2(g)$ is shown below.



- (i) Define the term *standard enthalpy change of formation*.

.....

.....

- (ii) Given that the enthalpy changes of the processes involved in the energy level diagram above are

- $\Delta H_f^\theta (PBr_3)$,
- $\Delta H_{vap}^\theta (PBr_3)$,
- bond energy (P–Br) and
- bond energy (Br–Br),

use them to assign to the enthalpy changes represented as

ΔH_1

ΔH_2

ΔH_4

ΔH_5

- (iii) Hence, use the information given below and relevant data from the *Data Booklet* to calculate the bond energy of P–Br bond.

ΔH_3	+314.6 kJ mol ⁻¹
ΔH_4	-184.5 kJ mol ⁻¹
ΔH_5	+38.8 kJ mol ⁻¹

bond energy (P–Br) = kJ mol⁻¹

[5]

[Total: 10]

- 4 Organic chemistry is the chemistry of carbon compounds. The types of organic reactions that you have studied are listed below.

addition
oxidation

substitution
reduction

elimination
hydrolysis

The natures of addition and substitution reactions are listed below and further described as follows,

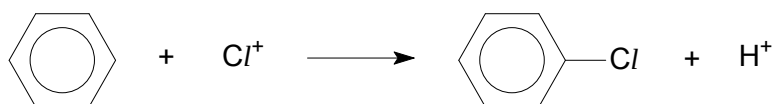
electrophilic

nucleophilic

free radical
(substitution only)

Electrophiles are electron-poor species that can form a covalent bond by accepting a pair of electrons from the reactive carbon. They can be either neutral or positively charged.

The chlorination of benzene is an example of an electrophilic substitution reaction. Cl^+ acts as an electrophile by accepting a pair of electrons from the reactive carbon to form the $\text{C}-\text{Cl}$ bond in this substitution reaction.



Conversely, nucleophiles are electron-rich species that can form a covalent bond by donating a pair of electrons to the reactive carbon. All nucleophiles possess a lone pair of electrons and they can be either neutral or negatively charged. An addition or substitution reaction involving a nucleophile is described as nucleophilic addition or nucleophilic substitution respectively.

A free radical contains an atom with an odd number of electrons in its valence shell and it is highly reactive. Examples of radicals include $\cdot\text{CH}_3$ and $\cdot\text{H}$.

- (a) Complete the table below by stating **only** the types of reaction using the lists given above. Use **both** lists given above where appropriate.

	organic reaction	type of reaction
(i)		
(ii)	$\text{CH}_3\text{CH}_2\text{CHBrCH}_3 \longrightarrow \text{CH}_3\text{CH}_2\text{CH(OH)CH}_3$	
(iii)		
(iv)		

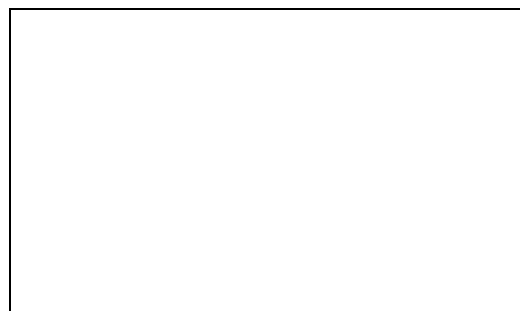
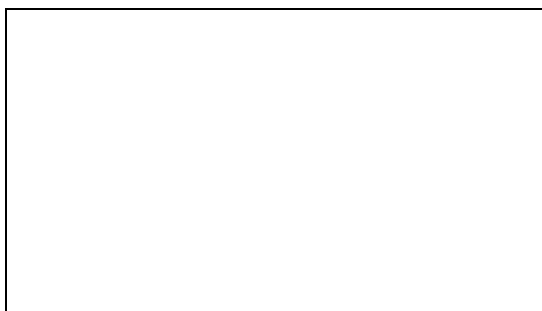
[6]

- (b) The reactant from (a)(i), ethylbenzene, can react with chlorine to undergo electrophilic substitution or free radical substitution depending on the reacting conditions.

State the two different conditions and draw two possible mono-chlorinated products that can be formed under each condition.

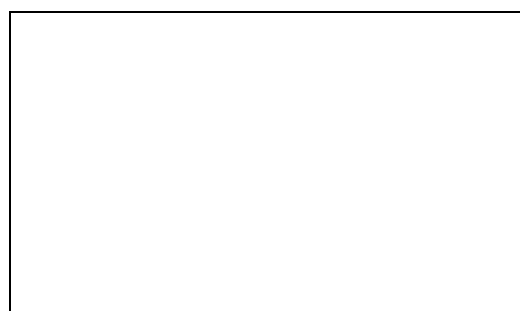
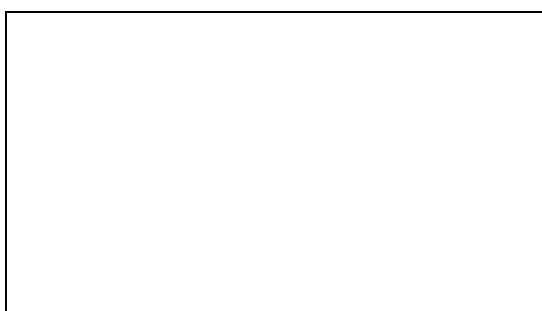
(i) Electrophilic Substitution

Condition:



(ii) Free Radical Substitution

Condition:



[4]
[Total: 10]

Section B

Answer **two** questions from this section on separate answer paper.

- 5 (a) (i) State and briefly explain the general trend of first ionisation energy from sodium to argon.
- (ii) With reference to relevant data from the *Data Booklet*, discuss the anomaly in the trend of first ionisation energy between phosphorous and sulfur.

[4]

- (b) Allotropy is the property of an element to exist in two or more different forms, known as allotropes e.g. diamond and graphite are allotropes of carbon.

One of the most common allotrope of phosphorous is white phosphorous, P_4 . It is produced industrially by heating phosphorous-containing rocks, fluorapatite, $Ca_5(PO_4)_3F$, to $1500\text{ }^\circ\text{C}$ with silicon dioxide and coke (carbon). The by-products of this process include carbon monoxide, calcium silicate ($CaSiO_3$) and calcium fluoride. The mole ratio of fluorapatite to white phosphorous produced is 4:3.

- (i) Write a balanced chemical equation for the process.
- (ii) Each atom of white phosphorous molecule has a valency of 3. Suggest by means of a diagram how the phosphorous atoms might be joined together.

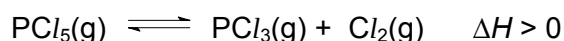
[2]

- (c) PCl_3 can be made in the laboratory by heating phosphorous with chlorine gas. In the presence of excess chlorine gas, PCl_5 will be produced.

Draw the structure of PCl_3 and PCl_5 and suggest the shape and bond angles of each molecule.

[4]

- (d) 10 g of PCl_5 vapour was heated in a closed container of 5.00 dm^3 flask at 500 K.



When equilibrium has been established, 45% of PCl_5 was dissociated.

- (i) Write an expression for K_c for this equilibrium, stating the units.
- (ii) Calculate the concentration, in mol dm^{-3} , of PCl_5 , PCl_3 and Cl_2 present at equilibrium.
- (iii) Calculate a value for K_c for the equilibrium.
- (iv) For each of the case below, deduce what will happen to the position of equilibrium when
- I volume of the container is halved;
- II temperature inside the flask is raised from 500 K to 600 K.

[10]

[Total: 20]

6 Compounds of sulfur have many uses in everyday life, e.g. disinfectants, matches and in making of paper.

(a) Phosphorus sulfide, P_4S_3 , is used in small amounts in the tip of a match stick. On striking a match stick, this compound burns to form sulfur dioxide and a solid oxide.

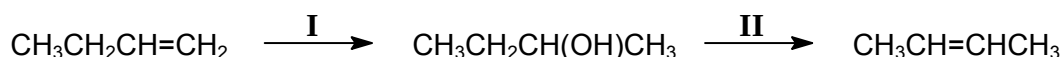
(i) Construct an equation for this reaction.

(ii) The melting points of the two oxides formed in (a)(i) differ significantly from that of silicon(IV) oxide (SiO_2).
Briefly relate this observation to the structure of, and bonding in, each of the oxides.

(iii) Both oxides formed in (a)(i) dissolve in water to give acidic solutions. Construct an equation for the reaction of each oxide with water and give an approximate pH for each of the solution formed.

[6]

(b) Concentrated sulfuric acid can be used under two different sets of reaction conditions to synthesise but-2-ene from but-1-ene via a 2-step reaction scheme as shown below.



(i) Describe the condition necessary for reaction II.

(ii) But-2-ene occurs in two isomeric forms but cyclobutene occurs in only one form.

Draw the displayed formulae of the two isomers of but-2-ene and explain why such isomerism is not possible with cyclobutene.

But-2-ene, on treatment with cold concentrated H_2SO_4 followed by the addition of hot water, gives an addition product **J**, $C_4H_{10}O$. **J** produces a yellow precipitate on warming with alkaline aqueous iodine.

J turns orange hot acidified potassium dichromate green. The product, **K**, formed is then reacted with HCN, in the presence of NaCN to give compound **L**, C_5H_9NO .

Treatment of **L** with hot, dilute sulfuric acid gives compound **M**. 0.590 g of **M** reacts with excess sodium metal to give 120 cm^3 of hydrogen gas (measured at room temperature and pressure).

On heating in the absence of air, **M** loses water to give a single compound **N** ($C_{10}H_{16}O_4$). **N** no longer reacts with sodium metal.

(iii) Deduce the structures of compounds **J**, **K**, **L**, **M** and **N** and use them to explain the chemistry of the reactions described.
[There is no need to comment on the chemistry of the formation of **J** from but-2-ene.]

(iv) An isomer of but-2-ene gives only one monobrominated product with Br_2 under uv light.

Suggest the structural formula of this isomer.

[14]

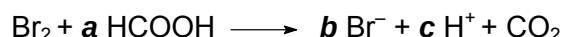
[Total: 20]

- 7 (a) Chlorine is found in the third period of the Periodic Table (sodium to argon). Some elements in the third period are solids, whilst some are gases at room temperature. This is attributed to the variation in melting points and boiling points of the elements across the third period.
- (i) State and explain the general trend in melting points of the elements in the third period of the Periodic Table (from sodium to argon) in terms of their structures and bonding.
- (ii) Sulfur, S_8 , exists as a yellow solid whilst chlorine, Cl_2 exists as a yellowish–green gas at room temperature and pressure. Suggest a reason for this difference.

[5]

- (b) Both bromine and chlorine can behave as oxidising agents.

Aqueous bromine oxidises methanoic acid, according the following equation.



- (i) Determine the values of **a**, **b** and **c** in the chemical equation.
- (ii) This reaction is catalysed by an acid. State and explain the effect of a catalyst on the rate constant for this reaction.

[3]

To investigate how the rate of reaction depends on the concentration of aqueous bromine and methanoic acid, two experiments were carried out.

Experiment 1: 20 cm³ of 0.25 mol dm⁻³ of aqueous bromine reacted with 20 cm³ of 5 mol dm⁻³ methanoic acid.

Experiment 2: 20 cm³ of 0.25 mol dm⁻³ of aqueous bromine reacted with 20 cm³ of 2.5 mol dm⁻³ methanoic acid.

The rate of the reaction was followed by measuring the volume of carbon dioxide gas produced at various times in the reaction. The results are tabulated below. All volumes are measured at room temperature and pressure (r.t.p).

Time / s	Volume of CO ₂ / cm ³	
	Experiment 1	Experiment 2
0	0	0
20	8.0	4.1
40	15.5	8.0
100	35.1	19.1
160	51.1	29.1
240	67.8	40.8
320	80.4	51.1
400	90.0	60.0
480	97.3	67.8
560	102.8	74.5

- (iii) Given that 1 mole of CO₂ occupies a volume of 24 000 cm³ at r.t.p, show that the maximum volume of CO₂ that can be produced from Experiment 1 is 120 cm³.

- (iv) Using the same axes, plot graphs of volume of CO_2 against time for Experiment 1 and 2. Label your graphs clearly.
- (v) Use your graph to determine the order of reaction with respect to Br_2 and HCOOH , showing your working clearly. Hence, give the rate equation for the reaction.
- (vi) In both experiments, the concentrations of HCOOH were far more concentrated than that of Br_2 . Explain the significance of doing this.
- (vii) Sketch a graph to show how the concentration of Br_2 in experiment 1 will vary with time.

Experiment 3 was carried out using $0.5 \text{ mol dm}^{-3} \text{ Br}_2(\text{aq})$ instead of 0.25 mol dm^{-3} while keeping the concentration of HCOOH at 5 mol dm^{-3} .

On the same axes, sketch another graph to illustrate how rate will vary in experiment 3. Label your graphs clearly.

[12]
[Total: 20]